

Week 2:

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Roll no: 2520030008

Week-2

The Librarian's Dashboard Scenario The library is now digital! The head librarian needs a report to see which members are active and which books are in the collection. You also need to handle a "Book Donation" process where a new book is added and assigned to a category at the exact same time—if one part fails, the whole process must stop to keep the catalog accurate.

Task to be Done

1. The Catalog Search (JOINS): Create a third table called Loans that connects member_id to book_id. Write a query to show the Member Name and the Book Title they have currently borrowed.
2. Collection Stats (Aggregations): Write a query to find the total number of books published in each year. o Hint: Use COUNT(book_id) and GROUP BY published_year.
3. The Donation Transaction (ACID): Use a Transaction (BEGIN...COMMIT) to add a new book to the Books table and simultaneously log the entry in a Donation_History table.
4. Search Speed (Indexing): Create an Index on the isbn column. Explain why searching by ISBN is now faster for the librarian than searching by Title. Skill Set Gained • Relational Logic: Connecting independent tables (Books + Members) through a Junction table (Loans). • Analytical SQL: Using GROUP BY to generate business insights. • Atomic Operations: Using Transactions to ensure multiple database changes succeed or fail as a single unit.

Solution:

Step 1:

```
1  /* Name: Sagi Sahasra Srihasa
2     Roll no: 2520030008
3  */
4
5
6
7  • CREATE DATABASE Donation_db;
8  • USE Donation_db;
9
10 • CREATE TABLE Books(
11     book_id INT PRIMARY KEY,
12     title VARCHAR(100) NOT NULL,
13     isbn VARCHAR(20) UNIQUE,
14     published_year INT CHECK(published_year < 2027)
15 );
16
17 • INSERT INTO Books VALUES
18 (1, 'The Great Gatsby', '9780743273565', 1925),
19 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
20 (3, '1984', '9780451524935', 1949);
21
22 • SELECT * FROM Books;
23
```

Output

Action Output

#	Time	Action	Message
✓ 1	12:58:59	CREATE DATABASE Donation_db	1 row(s) affected
✓ 2	12:59:01	USE Donation_db	0 row(s) affected
✓ 3	12:59:04	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) UNIQUE, publie...	0 row(s) affected
✓ 4	12:59:07	INSERT INTO Books VALUES (1, 'The Great Gatsby', '9780743273565', 1925), (2, 'To Kill a Mockingbird', '9780061120084', 1960), (3, '1984', '9780451524935', 1949);	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Step 2:

22 • `SELECT * FROM Books;`

Result Grid

	book_id	title	isbn	published_year
▶	1	The Great Gatsby	9780743273565	1925
	2	To Kill a Mockingbird	9780061120084	1960
	3	1984	9780451524935	1949
•	NULL	NULL	NULL	NULL

Books 1 x

Output

Action Output

#	Time	Action
✓ 1	12:58:59	CREATE DATABASE Donation_db
✓ 2	12:59:01	USE Donation_db
✓ 3	12:59:04	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) UNI
✓ 4	12:59:07	INSERT INTO Books VALUES (1,'The Great Gatsby','9780743273565',1925), (2,'To Kill a Mockingbird','9780061120084
✓ 5	13:00:10	SELECT * FROM Books LIMIT 0, 1000

Step 3:

24 • `CREATE TABLE Members(`

25 `member_id INT PRIMARY KEY,`

26 `full_name VARCHAR(100),`

27 `email VARCHAR(100) UNIQUE`

28 `);`

29

30 • `INSERT INTO Members VALUES`

31 `(101,'Sahasra','sahasra@gmail.com'),`

32 `(102,'Bhanu','bhanu@gmail.com'),`

33 `(103,'Nikshitha','nikshi@gmail.com');`

34

Output

Action Output

#	Time	Action	Mess
✓ 2	12:59:01	USE Donation_db	0 row
✓ 3	12:59:04	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) UNIQUE, pu...	0 row
✓ 4	12:59:07	INSERT INTO Books VALUES (1,'The Great Gatsby','9780743273565',1925), (2,'To Kill a Mockingbird','9780061120084',1960), (3,'...	3 row
✓ 5	13:00:10	SELECT * FROM Books LIMIT 0, 1000	3 row
✓ 6	13:00:52	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE)	0 row
✓ 7	13:00:55	INSERT INTO Members VALUES (101,'Sahasra','sahasra@gmail.com'), (102,'Bhanu','bhanu@gmail.com'), (103,'Nikshitha','nikshi@g...	3 row

Step 4:

```
37 • SELECT * FROM Members;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	member_id	full_name	email
▶	101	Sahasra	sahasra@gmail.com
	102	Bhanu	bhanu@gmail.com
	103	Nikshitha	nikshi@gmail.com
*	NULL	NULL	NULL

Members 2 x

Output:

Action Output ▾

	#	Time	Action
✔	3	12:59:04	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100)
✔	4	12:59:07	INSERT INTO Books VALUES (1,'The Great Gatsby','9780743273565',1925), (2,
✔	5	13:00:10	SELECT * FROM Books LIMIT 0, 1000
✔	6	13:00:52	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARI
✔	7	13:00:55	INSERT INTO Members VALUES (101,'Sahasra','sahasra@gmail.com'), (102,'Bh
✔	8	13:02:03	SELECT * FROM Members LIMIT 0, 1000

Step 5:

```
39 • CREATE TABLE Loans(  
40     loan_id INT PRIMARY KEY,  
41     member_id INT,  
42     book_id INT,  
43     loan_date DATE,  
44     FOREIGN KEY(member_id) REFERENCES Members(member_id),  
45     FOREIGN KEY(book_id) REFERENCES Books(book_id)  
46 );  
47  
48 • INSERT INTO Loans VALUES  
49     (1,101,1,'2025-01-05'),  
50     (2,102,2,'2025-01-08'),  
51     (3,103,3,'2025-01-10'),  
52     (4,101,2,'2025-02-01'),  
53     (5,102,1,'2025-02-05'),  
54     (6,103,2,'2025-02-12'),  
55     (7,101,3,'2025-03-01'),  
56     (8,102,3,'2025-03-07'),  
57     (9,103,1,'2025-03-15'),  
58     (10,101,1,'2025-04-01');  
59
```

✓	5	13:00:10	SELECT * FROM Books LIMIT 0, 1000
✓	6	13:00:52	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE)
✓	7	13:00:55	INSERT INTO Members VALUES (101,'Sahasra','sahasra@gmail.com'), (102,'Bhanu','bhanu@gmail.com'), (103,'Nikshitha','nikshi
✓	8	13:02:03	SELECT * FROM Members LIMIT 0, 1000
✓	9	13:02:35	CREATE TABLE Loans(loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN KE
✓	10	13:02:40	INSERT INTO Loans VALUES (1,101,1,'2025-01-05'), (2,102,2,'2025-01-08'), (3,103,3,'2025-01-10'), (4,101,2,'2025-02-01'), (5,10

Step 6:

60 • `SELECT * FROM Loans;`

61

Result Grid

	loan_id	member_id	book_id	loan_date
▶	1	101	1	2025-01-05
	2	102	2	2025-01-08
	3	103	3	2025-01-10
	4	101	2	2025-02-01
	5	102	1	2025-02-05
	6	103	2	2025-02-12
	7	101	3	2025-03-01
	8	102	3	2025-03-07
	9	103	1	2025-03-15
	10	101	1	2025-04-01
*	NULL	NULL	NULL	NULL

Loans 3 x

Output

Action Output

#	Time	Action	
✓	6	13:00:52	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100),
✓	7	13:00:55	INSERT INTO Members VALUES (101,'Sahasra','sahasra@gmail.com'), (102,'Bhanu','bhanu@
✓	8	13:02:03	SELECT * FROM Members LIMIT 0, 1000
✓	9	13:02:35	CREATE TABLE Loans(loan_id INT PRIMARY KEY, member_id INT, book_id INT,
✓	10	13:02:40	INSERT INTO Loans VALUES (1,101,1,'2025-01-05'), (2,102,2,'2025-01-08'), (3,103,3,'2025-0
✓	11	13:03:42	SELECT * FROM Loans LIMIT 0, 1000

Step 7:

```

62 • SELECT
63     m.full_name AS Member_Name,
64     b.title AS Book_Title
65 FROM Loans l
66 INNER JOIN Members m
67 ON l.member_id = m.member_id
68 INNER JOIN Books b
69 ON l.book_id = b.book_id;
70

```

	Member_Name	Book_Title
▶	Sahasra	The Great Gatsby
	Sahasra	To Kill a Mockingbird
	Sahasra	1984
	Sahasra	The Great Gatsby
	Bhanu	To Kill a Mockingbird
	Bhanu	The Great Gatsby
	Bhanu	1984
	Nikshitha	1984
	Nikshitha	To Kill a Mockingbird
	Nikshitha	The Great Gatsby

Result 4 x

Output

Action Output

#	Time	Action
✓	7 13:00:55	INSERT INTO Members VALUES (101,'Sahasra','sahasra@gmail.com'), (102,'Bhanu','bl
✓	8 13:02:03	SELECT * FROM Members LIMIT 0, 1000
✓	9 13:02:35	CREATE TABLE Loans(loan_id INT PRIMARY KEY, member_id INT, book_id I
✓	10 13:02:40	INSERT INTO Loans VALUES (1,101,1,'2025-01-05'), (2,102,2,'2025-01-08'), (3,103,3,'
✓	11 13:03:42	SELECT * FROM Loans LIMIT 0, 1000
✓	12 13:04:04	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans I INN

Step 8:

```

71 • SELECT
72     published_year,
73     COUNT(book_id) AS Total_Books
74 FROM Books
75 GROUP BY published_year;
76

```

	published_year	Total_Books
▶	1925	1
	1960	1
	1949	1

✓ 13 13:04:36 SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year LIMIT 0, 10

Step 9:

```
77 • CREATE TABLE Donation_History(  
78     donation_id INT PRIMARY KEY,  
79     book_id INT,  
80     donor_name VARCHAR(100),  
81     donation_date DATE,  
82     FOREIGN KEY(book_id) REFERENCES Books(book_id)  
83 );  
84
```

```
85 • START TRANSACTION;  
86
```

```
87 • INSERT INTO Books VALUES  
88     (4,'Animal Farm','9780451526342',1945);  
89
```

```
90 • INSERT INTO Donation_History VALUES  
91     (1,4,'Raj Kumar',CURDATE());  
92
```

```
93 • COMMIT;  
94
```

Output

Action Output

#	Time	Action
14	13:05:38	CREATE TABLE Donation_History(donation_id INT PRIMARY KEY, book_id INT, donor_n
15	13:05:45	START TRANSACTION
16	13:05:48	INSERT INTO Books VALUES (4,'Animal Farm','9780451526342',1945)
17	13:05:50	INSERT INTO Donation_History VALUES (1,4,'Raj Kumar',CURDATE())
18	13:05:55	COMMIT
19	13:06:01	CREATE INDEX idx_isbn ON Books(isbn)

Step 10:

```
101 • SELECT *  
102 FROM Books  
103 WHERE isbn='9780451524935';
```

Result Grid



Filter Rows:

Edit:



Export

	book_id	title	isbn	published_year
▶	3	1984	9780451524935	1949
*	NULL	NULL	NULL	NULL

```
20 13:07:10 SELECT * FROM Books WHERE isbn='9780451524935' LIMIT 0, 1000
```